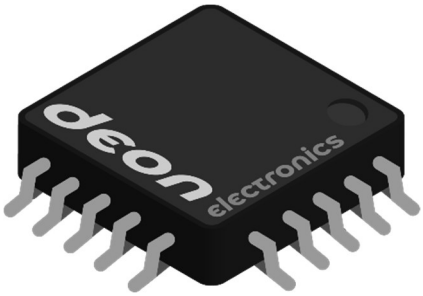

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CRITICAL ITEM DEVELOMENT SPECIFICATION (CIDS) For BlinkerOne

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Entity Identifier	Document Identifier	Document Type Code	Revision
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CIDS DOCUMENTARY CONTROL

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1. FIELD OF APPLICATION

1.1. IDENTIFICATION

This specification defines the requirements concerning the functional and technical characteristics, the performances and the design constraints for the following HW item:

Name of HW item:	Blinker One
Identification number:	001
Acronym:	(BlinkerOne)

Table 1-1 Identification

:

1.2. HW ITEM OVERVIEW

This section describes the HW ITEM functional architecture, its interfaces, and its operational role. This information is not a requirement but aims to introduce the requirements provided in the following sections of this CIDS. Figure below provides a functional synopsis of the Interface HW item. The specified item is an Open-Source Console & Versatile Companion.

(BlinkerOne) is an open source handheld console based on the ESP32-S3, designed as a versatile tool:

- Retro game emulation (old consoles, simple homebrew games)
- Custom virtual environments or small DIY “OS-like” systems
- Home automation / IoT control via Wi-Fi and Bluetooth
- Experiments in UI/UX, haptics, and embedded audio

The goal is not just to build a “modern Tamagotchi”, but a modular platform to hack, learn, and prototype interactive physical interfaces.

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1.3. DOCUMENT PURPOSE

The purpose of this document is to specify all the requirements relating to the (BlinkerOne) HW item. All the operating or maintenance requirements which may affect the design or the manufacturing process should also be specified in this document. Once validated by a dedicated VTR review (Validation Technical Review), it will be the reference document for the design and verification of the HW item. The following figure provides an overview of the project documentary architecture. This figure is provided for information only to facilitate the understanding of the document, as the official reference of the documentary organization is described in the project management plan.

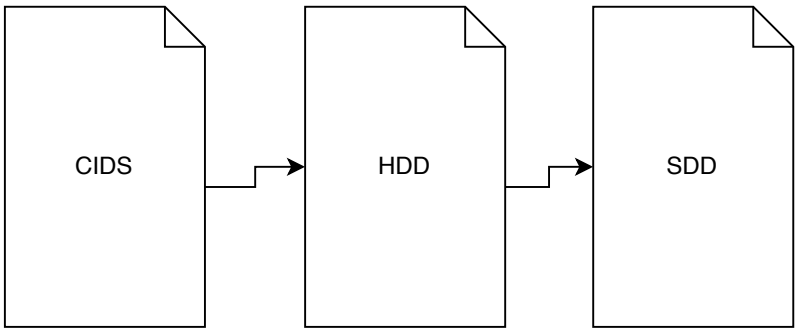

Figure 1-1 [BlinkerOne] Documentary architecture

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2. REFERENCED DOCUMENTS

This chapter is not part of the formal specification: the documents listed are given by way of reference, meaning that if formal conformity has to be proved, a specific requirement is drawn up stipulating the specific part of the document the system has to comply with. If "Revision – Date" is stipulated in the tables below, this revision should be taken into account. If not, the revision to be taken into account is the most recent revision at the time this document is issued.

2.1. Applicable DOCUMENTS

2.1.1. Normative documents

[Callout]	Reference	Title	Revision - Date	Origin
Directive	2011/65/EU & 2015/863	Restriction of Hazardous Substances (RoHS 3)	Rev 2015	European Union
Directive	EC 1907/2006	Registration, Evaluation, Authorisation and Restriction of Chemicals	Latest SVHC List	European Union / ECHA
Directive	IPC-A-610	Acceptability of Electronic Assemblies	Rev. H (2020)	IPC

Table 2-1 Normative documents

2.1.2. Input documents

[Callout]	Reference	Title	Revision - Date	Origin
Datasheet	ESP32-S3-WROOM-1	ESP32-S3-WROOM-1	Rev 1.8	Espressif Systems
Datasheet	MPU-6050	MPU-6050 Six-Axis (Gyro + Accelerometer) MEMS	Rev 3.4	TDK InvenSense
Datasheet	MAX98357A	Class D Amplifier with Class AB Performance	Rev 11	Maxim Integrated
Datasheet	DRV2605L	DRV2605L Haptic Driver for ERM and LRA	Rev MARCH 2018	Texas Instruments

Table 2-2 input documents

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3. REQUIREMENTS

3.1. Item Definition

This section briefly describes the overall architecture.

- Description: The electronic board shall act as a standalone embedded system capable of processing user inputs (buttons, motion sensors), acquiring audio, storing data, and providing haptic feedback. Core

- Component: Main control, data processing, and wireless communications (Wi-Fi/Bluetooth) shall be handled by an ESP32-S3-WROOM-1-N16R8 module.

3.2. Functional and Performance Requirements

Here, we detail the expectations for each hardware block of your board.

3.2.1. User interfaces (Discrete Inputs)

- The board shall integrate a directional pad (D-pad).
- The board shall integrate 2 action buttons (digital switches)
- ***Constraint:*** Signals from these buttons must be filtered (hardware or software debounce) to ensure a response time of less than 50 ms.

3.2.2. Motion sensor

- An MPU-6050 gyroscope/accelerometer (6-axis) shall be integrated via an I2C bus.
- It must detect the board's orientation and acceleration with a refresh rate suitable for the application.

3.2.3. Haptic Feedback

- A Linear Resonant Actuator (LRA) motor shall be driven by the board to provide tactile feedback to the user.
- ***Constraint:*** It will require a suitable driver (e.g., I2C haptic driver or specific PWM signal) to generate precise vibration profiles.

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3.2.4. Audio and Storage

- An Analog-to-Digital Converter (I2S ADC) shall be present for audio acquisition.
- The board shall integrate an SD card reader for local storage (specify the interface: SPI or SDMMC for better performance).

3.2.5. Connectivity and Power

- A USB-C interface shall be integrated.
- It will serve both for data transmission (flashing the ESP32, serial communication) and for powering the board.
- WiFi and Bluetooth (for many application like HomeAssistant integration, etc…)

3.3. Electrical Requirments

- The system shall be powered via the USB-C port (5V).

- The board must integrate the necessary voltage regulators (notably a robust 3.3V LDO to stably power the ESP32, MPU-6050, and SD card).

- Attention point: Power supply sizing must account for current peaks from the LRA motor and the ESP32 (especially during Wi-Fi transmission).

3.4. Mechanical and Environmental Requirements

- Form factor: Maximum PCB dimensions (approx.: 100mm x 40mm).

- Ergonomics: The placement of the buttons, D-pad, and triggers must comply with the mechanical design of the enclosure (if applicable).

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